THE CHARACTER OF A BORCHERDS FORM ON A QUOTIENT OF POSITIVE GENUS, AND THE REDUCTION TYPE OF A GENUS-ONE QUOTIENT

1. Purpose

Guo and Yang [1] construct equations of Shimura curves $X_0^D(N)/W$ from Borcherds forms whose values at CM points are given by Schofer's formula. When the quotient has genus zero the construction is unconditional. For a quotient of positive genus it rests on their Assumption 40, which says that a Borcherds form satisfying the criterion of their Lemma 7 has trivial character. This note records what that assumption amounts to, when it holds, a case in which it fails, how to do without it, and a way to determine which curve of an isogeny class a genus-one quotient is that uses nothing of the Borcherds construction.

Throughout, $X = X_0^D(N)/W$ is a quotient of a Shimura curve by a group W of Atkin–Lehner involutions, Γ is the Fuchsian group with $X(\mathbb{C}) = \Gamma\backslash\mathcal{H}$, and ψ is a Borcherds form on X of weight 0 attached to a weakly holomorphic form F with integral principal part, so that ψ is a meromorphic function on $\mathcal{H}$ with $\psi(\gamma z) = \chi(\gamma)\,\psi(z)$ for a character χ of Γ, and $D = \operatorname{div}(\psi)$ is a $\mathbb{Q}$-rational divisor of degree 0 on X supported on CM points. The character χ has finite order, say n.

2. What Assumption 40 says

The group Γ is generated by hyperbolic elements $\alpha_1, \beta_1, \dots, \alpha_g, \beta_g$ and elliptic elements $\gamma_1, \dots, \gamma_r$ of orders $k_1, \dots, k_r$, subject to $\prod_j [\alpha_j, \beta_j] \prod_i \gamma_i = 1$ and $\gamma_i^{k_i} = 1$. Lemma 7 of [1] (Lemma 8 of [2]) says: if $c_0(0)$ is even, then $\chi(\gamma_i) = 1$ for every i with $k_i \neq 3$ if and only if, at each elliptic point of order $k_i \neq 3$, the order of ψ has the parity of $c_0(0)/2$. This is a condition on the divisor D alone. When $g = 0$ there are no hyperbolic generators, so it decides χ; this is why the equations in the main table of [1] are unconditional. Assumption 40 is the statement that the same parity condition decides χ when $g > 0$; as Remark 41 of [1] says, what is assumed is $\chi(\alpha_j) = \chi(\beta_j) = 1$.

3. The character is trivial exactly when the divisor is principal

Proposition 3.1. *Suppose the parity condition of Lemma 7 holds, so that χ is trivial on the elliptic generators. Then the following are equivalent:*

1. *χ is trivial, i.e. ψ is a function on X;*
2. *D is a principal divisor on X;*
3. *the class of D vanishes in $\operatorname{Jac}(X)(\mathbb{Q})_{\mathrm{tors}}$.*

In general the order of χ is the order of the class of D in $\operatorname{Jac}(X)(\mathbb{Q})$.

Proof. Since χ has order n, ψ^n is invariant under Γ, hence a function on X with divisor nD; so the class of D is torsion, and it is a $\mathbb{Q}$-rational class since D is $\mathbb{Q}$-rational. If $D = \operatorname{div}(h)$ for a function h on X, then $u = \psi/h$ is a holomorphic function on $\mathcal{H}$ without zeros or poles with $u(\gamma z) = \chi(\gamma)u(z)$. Since $|\chi| = 1$, $\log|u|$ is a Γ-invariant harmonic function on $\mathcal{H}$, hence a harmonic function on the compact Riemann surface X (bounded near the elliptic points, so it extends across them), hence constant. A holomorphic function of constant modulus is

constant, so u is constant and $\chi = 1$. Conversely if $\chi = 1$ then $D = \mathrm{div}(\psi)$ is principal. The last sentence follows by applying the argument to ψ^m for $m \mid n$. □

Corollary 3.2. *If the degree-zero CM divisors of X generate a subgroup of $\mathrm{Jac}(X)(\mathbb{Q})$ with no torsion, in particular if X has genus one and $E = \mathrm{Jac}(X)$ has $E(\mathbb{Q})_{\mathrm{tors}} = 0$, then every Borcherds form on X satisfying the parity condition has trivial character: Assumption 40 holds for every F. In general the order of χ divides the exponent of $\mathrm{Jac}(X)(\mathbb{Q})_{\mathrm{tors}}$.*

So Assumption 40 is unconditional for $X_0^{142}(1)^*$, $X_0^{158}(1)^*$ and $X_0^{38}(5)^*$, whose Jacobians $142a1$, $158a1$ and $190a1$ have no rational torsion, and it cannot be decided from F alone when there is rational torsion: the class of D in $\mathrm{Jac}(X)(\mathbb{Q})_{\mathrm{tors}}$ is not a function of the discriminants and multiplicities in D.

Remark 3.3 (How a failure shows)**.** Schofer's formula gives $\log|\psi|$ at a CM point, and $|\psi|$ is well defined on X whatever χ is, since $|\chi| = 1$. If χ has order n then $|\psi|^n = |\psi^n|$ is the absolute value of a rational function evaluated at a rational point, hence rational when everything is normalised over $\mathbb{Q}$, and $|\psi|$ is an n-th root of a rational number. A value such as $2 \cdot 7^{3/2} \cdot 17^{1/2}$ is therefore the signature of a character of order 2.

Example 3.4 (Assumption 40 fails at $X_0^{14}(17)^*$)**.** The star curve $X = X_0^{14}(17)/W_{14,17}$ has genus one and lies in the isogeny class $238a$, whose two curves have $E(\mathbb{Q}) \cong \mathbb{Z}/2 \times \mathbb{Z}$. The genus-one star driver of the repository (`tools/genus1star.m`) found a Borcherds form on X whose value at the CM point of discriminant -4 is $2 \cdot 7^{3/2} \cdot 17^{1/2}$ (the exponents $3/2$ and $1/2$ sit at the primes $7 \mid D$ and $17 \mid N$). By Remark 3.3 its character has order 2: the divisor of that form is a nonzero element of $E(\mathbb{Q})[2]$. This is a case where the parity condition of Lemma 7 holds and Assumption 40 is false. (A computed value; the form is in the run log cited in the driver.)

4. Doing without the assumption

Let e be the exponent of $\mathrm{Jac}(X)(\mathbb{Q})_{\mathrm{tors}}$ (for a genus-one quotient, of $E(\mathbb{Q})_{\mathrm{tors}}$, taken over the curves of the isogeny class when the curve is not yet known). By Proposition 3.1, ψ^e is a function on X for every Borcherds form ψ satisfying the parity condition, with divisor eD, and Schofer's formula gives $|\psi^e| = |\psi|^e$ exactly. Building the coordinates of [1], §4.3 from ψ^e instead of ψ therefore needs no assumption: the coordinates have degree e times what they would otherwise have, the relation between them has correspondingly higher bidegree, and nothing else changes. When $E(\mathbb{Q})$ is torsion-free, $e = 1$. The repository's driver takes this route on quotients with rational torsion.

5. The reduction type of a genus-one quotient from the dual graph

The curve inside the isogeny class can be read off the Čerednik–Drinfeld model, with nothing of the Borcherds construction involved. Let $p \mid D$. The special fibre of the stable model of X over $\mathbb{Z}_p^{\mathrm{nr}}$ is a union of rational curves meeting at nodes, and its dual graph with each edge weighted by the thickness of the node is the quotient by W of the dual graph of $X_0^D(N)$, with the length of an edge the order of the stabiliser of the edge in the group acting (this is the metric refinement of the dual-graph filter of the repository; the lengths computed by `DualGraphMetricQuotientFromData` are stabiliser orders in the group containing the centre $\{\pm 1\}$, hence twice the thickness).

Proposition 5.1. *Let X have genus one with Jacobian E, and let $p \mid D$. Then E has multiplicative reduction at p of Kodaira type I_n, where n is the sum of the thicknesses of the nodes*

of the stable model of X at p, equivalently half the total length of the metric dual graph after contracting its trees.

Proof. The minimal regular model of X over $\mathbb{Z}_p^{\mathrm{nr}}$ is obtained from the stable model by replacing each node of thickness t by a chain of $t-1$ rational curves. Its special fibre is connected of arithmetic genus one, with every component rational, so its dual graph is a cycle; contracting the trees of the metric dual graph of the stable model leaves a cycle whose total length is the number of components of that special fibre, which for a curve of genus one with multiplicative reduction is the index n of I_n. (Over $\mathbb{Q}_p$ the type may be nonsplit, but n does not change under unramified base change.) □

Computation 5.2. `KodairaIndexFromDualGraph` computes n; `tests/KodairaFromDualGraph.m` checks it. Calibration on the three stars whose curve is known (the newform's isogeny class has a single curve): all six indices agree with Cremona's tables. Identification of the two stars known only up to isogeny from the Borcherds construction:

quotient	dual graph	curves of the class	curve
$X_0^{142}(1)^*$	I_9 at 2, I_1 at 71	$142a1$: I_9, I_1	$142a1$
$X_0^{158}(1)^*$	I_8 at 2, I_1 at 79	$158a1$: I_8, I_1	$158a1$
$X_0^{38}(5)^*$	I_{11} at 2, I_1 at 19	$190a1$: I_{11}, I_1	$190a1$
$X_0^{14}(17)^*$	I_7 at 2, I_4 at 7	$238a1$: I_{14}, I_2; $238a2$: I_7, I_4	$238a2$
$X_0^{94}(3)^*$	I_8 at 2, I_1 at 47	$282b1$: I_8, I_1; $282b2$: I_4, I_2	$282b1$

Both identifications agree with the curves the Borcherds construction produced under Assumption 40, including at $X_0^{14}(17)^*$ where the assumption fails for some forms and the model was built from others. The heights of the branch points on the quotient curve give a third, independent identification: on the wrong curve of the class the predicted height is not the height of a point of the Mordell–Weil lattice.

References

[1] J.-W. Guo and Y. Yang, Equations of hyperelliptic Shimura curves, Compositio Math. 153 (2017), 1–40.
[2] Y. Yang, Schofer's formula and the Borcherds forms on Shimura curves, 2015.